Dataset: The dataset used in this project is derived from the **Twitter US Airline Sentiment** dataset, which collects tweets related to major U.S. airline companies for a total of **10,445 tweets**.

General idea

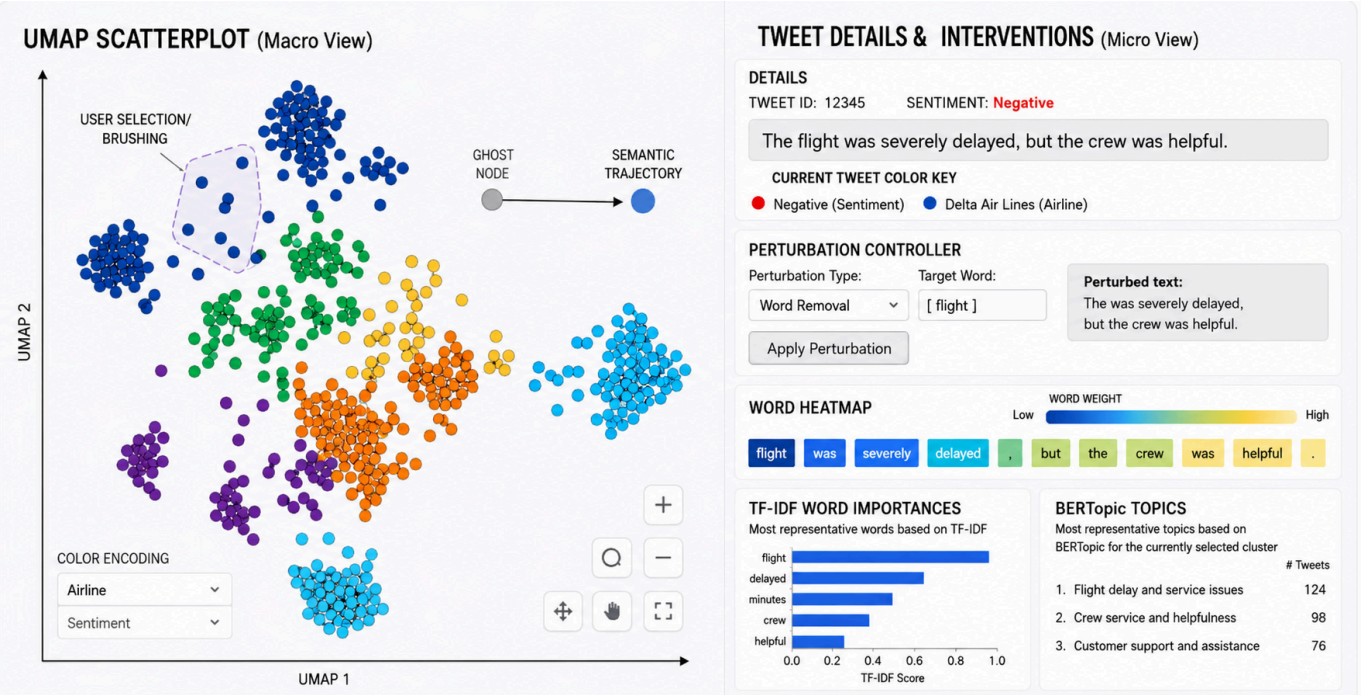
Analytics: The primary goal is to evaluate an embedding model like the **Qwen3-Embedding-0.6B** model by testing its semantic robustness through controlled textual perturbations. Each tweet is encoded into a high-dimensional embedding that captures its semantic meaning. To test how sensitive this embedding space is to structural and semantic changes we apply a set of perturbation operations.
Visual: The system provides an interactive 2D representation of the tweets. Each tweet is displayed as a node in the scatterplot. When transformations are applied, the system visually maps the semantic drift using real-time animated trajectories. A ghost node remains at the original coordinate, while an arrow traces the path to the newly generated embedding coordinate.

Intended users: The intended users of this system are researchers and students in Natural Language Processing (NLP) and data scientists, who need to interpret latent embedding spaces.

Visual Analytics Cycle:

- **Data:** Raw tweets are transformed into high-dimensional vectors by the embedding model.
- **Models:** A dimensionality reduction technique (e.g., PCA or UMAP) is applied to project embeddings into a 2D space. Furthermore, the system integrates BERTopic to compute the main latent topics across the entire dataset, while TF-IDF is used to extract the most distinctive keywords for user-selected clusters of tweets. User interactions trigger real-time computations: the Gemma-4 model is used to compute the tweet paraphrases, and cosine similarity algorithms calculate the impact of semantic shifts for individual words.
- **Visualization:** The dashboard consists of multiple coordinated views: a scatterplot, a word-heatmap**,** a horizontal bar chart (TF-IDF) and a semantic topic panel (BERTopic). Selecting a single tweet on the scatterplot triggers the rendering of the diagnostic heatmap for that specific text, whereas applying a perturbation to a specific word within the heatmap updates the scatterplot to display the resulting semantic drift. Furthermore, selecting multiple tweets via spatial brushing dynamically generates the TF-IDF bar chart and identifies the dominant BERTopic for that specific area. Additionally, selecting a keyword bar in the TF-IDF graph highlights the tweets containing that word in the scatterplot. Similarly, interacting with a semantic topic in the BERTopic panel highlights the subset of tweets belonging to that specific latent theme**.**
- **Knowledge:** Users can visually infer which words anchor a sentence's meaning and identify potential blind spots or biases in the embedding model. Furthermore, they can understand the impact of different perturbation strategies like generative paraphrasing or simple dictionary swaps that produce semantic shifts in the latent space.
- **Feedback Loop:** The system supports an iterative exploration process where the user explores semantic clusters, applies textual transformations, observes the visual changes and refines the analysis by trying new modifications.

Mockup



Participants

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